Supplementary Methods

S1. Waveform features and classifier

Features. For every 2-s window of the 500 Hz waveform we computed: - the fraction of finite samples; - the mean, minimum, maximum, 2nd, 10th, 90th and 98th centiles; - the median absolute second difference; - the number of systolic upstrokes in the window and in the centred 6-s window.

An upstroke is a rise of more than 8 mmHg within 100 ms after 20-ms smoothing; successive upstrokes are at least 250 ms apart.

Labels. The labels, thresholds and guard margins are given in the Methods.

Cases without a waveform. Of 3,645 cases with a listed waveform track, 211 returned no waveform through the VitalDB API on 26 October 2026. Their case identifiers are released with the code.

Threshold tuning. The noise threshold, the upstroke rule and the guard length were set by inspecting outcome-free descriptive data from fewer than 600 cases during development. The rule discarded before the freeze (beat-lag autocorrelation) labelled 12% of windows as non-pulsatile in patients with atrial fibrillation against 1% in sinus rhythm, at pulse pressures of 56–83 mmHg.

S2. Interval for the ratio of odds ratios (amendment A15)

Why the bootstrap failed. Exposure categories under F1 × R3 and the reference differed in 33 of 2,563 patients. Bootstrap resamples change few of these patients, so the bootstrap distribution of log ROR is spiky and its central 95% is narrow. In 100 permutations of the outcome, under which the true ROR is exactly 1, the percentile interval covered 1 in 51% of datasets for F1 × R3, 79% for F0 × R1, 89% for F3 × R2 and 91% for F6 × R3.

The rule. Before the real outcome was examined, a written rule set every ROR-type interval to log ROR ± z × max(bootstrap SE, permutation-null SD). The permutation-null SD came from 200 permutations of the analysed outcome. The same SE was used for the simultaneous bands, whose critical value is the bootstrap max-|z| over the 35 cells (3.22).

Coverage check. The SD was estimated from 200 permutations in the dry run, independent of the 100 calibration datasets. With that SD, the interval covered 1 in 100%, 99%, 100% and 97% of calibration datasets for the four pairs.

Operating characteristics at the margin. Outcomes were simulated from the fully adjusted reference model, adding a log-odds shift γ in the patients moved upwards by the pipeline. We chose γ = 1.91 so that the median simulated ROR was 1.24. In 200 datasets, each analysed with the full rule (200 bootstrap resamples and 200 permutations), equivalence was declared in 5%. Without differential error (γ = 0) it was declared in 82%, and the interval covered 1 in 97%.

Reaching the lower margin requires almost all five downward-moved patients to develop AKI. The SD of log ROR under the simulated alternative (0.098) exceeded that under the null (0.062). The bootstrap SE, which grows with the number of cases among the discordant patients, then dominates the rule.

S3. Blinded AI rating

The 400-snippet adjudication set has five strata: - 100 low-MAP windows flagged as artefact; - 50 low-MAP windows labelled guard only; - 100 low-MAP windows labelled valid; - 100 random valid windows; - 50 flagged windows with MAP ≥ 65 mmHg.

There are at most two snippets per patient. A subset of 152 snippets was shown, as 26 grids of six 40-s waveform panels, to two independent instances of an AI model (Claude, Anthropic). The grids were in a folder that contained no key, code or labels. The raters could read files, were instructed to open only the listed images, and saw only the waveform.

The prompt is released with the code. The subset is enriched for artefact (43% against 3.7% in the data), which inflates κ.

S4. Injection–recovery

The implementation check used 296 clean 2-min segments from 100 cases. Synthetic flush, zeroing, sampling, damping, noise and offset artefacts, after Li et al., were injected in the middle 20 s.

The post hoc sweep used 119 segments from 60 cases. It varied: - flush amplitude (120–300 mmHg) and duration (0.5–4 s); - open-transducer level (0–25 mmHg); - damping cut-off (1.5–10 Hz); - partial occlusion (pulse pressure scaled to 0.5–0.15); - linear drift (10–40 mmHg over 20 s); - 18-Hz ringing (5–20 mmHg); - sampling without a terminal flush.

Detection counted only artefact labels, not guard. Clean segments were selected as valid by the classifier, so the absence of false labels in uninjected segments holds by construction.

S5. Real-time episode table

For each episode and each elapsed time t (10, 30, 60 and 120 s), the nadir so far was the lowest monitor MAP in the first t seconds. Episodes shorter than t were not counted at t. The table therefore gives the probability that an episode is flagged, given that it has lasted at least t seconds and reached a given nadir.

Supplementary tables

Table S1. Recorded minutes with MAP < 65 mmHg classified against the reference (descriptive cohort)

res	filter	low_min	spurious_fra	lo	hi	confirmed_bridged_low	indeterminate	n_patients_with_spurious
R1	F0	6.16e+04	0.023	0.021	0.025	0.932	0.012	0.032	2196
R1	F1	6.09e+04	0.020	0.019	0.022	0.943	0.012	0.025	2104
R1	F2	6.06e+04	0.018	0.017	0.020	0.947	0.011	0.024	2060
R1	F3	5.84e+04	0.006	0.005	0.006	0.974	0.008	0.012	1921
R1	F4	5.96e+04	0.009	0.008	0.009	0.963	0.009	0.020	2041
R1	F5	5.96e+04	0.009	0.008	0.009	0.963	0.009	0.020	2029
R1	F6	5.91e+04	0.030	0.028	0.032	0.942	0.010	0.019	2683
R2	F0	6.22e+04	0.034	0.031	0.037	0.934	0.019	0.013	1108
R2	F1	6.19e+04	0.033	0.030	0.036	0.938	0.019	0.010	1065
R2	F2	6.2e+04	0.033	0.030	0.036	0.938	0.020	0.010	1081
R2	F3	6.13e+04	0.031	0.029	0.034	0.946	0.019	0.004	1079
R2	F4	6.2e+04	0.033	0.030	0.036	0.938	0.020	0.009	1098
R2	F5	6.09e+04	0.018	0.017	0.020	0.953	0.019	0.009	754
R2	F6	5.15e+04	0.099	0.094	0.105	0.868	0.023	0.010	1338
R3	F0	5.96e+04	0.008	0.007	0.009	0.960	0.006	0.027	363
R3	F1	5.93e+04	0.007	0.006	0.007	0.964	0.006	0.023	311
R3	F2	5.94e+04	0.005	0.005	0.006	0.963	0.008	0.024	271
R3	F3	5.88e+04	0.003	0.003	0.004	0.970	0.009	0.018	170
R3	F4	5.92e+04	0.004	0.003	0.004	0.966	0.008	0.023	195
R3	F5	5.92e+04	0.003	0.003	0.004	0.967	0.008	0.022	177
R3	F6	5.19e+04	0.075	0.071	0.080	0.887	0.011	0.027	1105

low_min: pipeline minutes < 65; fractions are of low_min; lo/hi: patient-cluster bootstrap 95% CI of the spurious fraction.

Table S2. Agreement of the binary exposure (≥ 10 min MAP < 65) with the reference (descriptive cohort)

filter	res	prev_ref	prev_pipe	sens	spec	kappa	flips	up	down	mean_diff	loa_lo	loa_hi
F0	R1	0.423	0.441	0.999	0.967	0.960	64	62	2	0.859	-6.257	7.975
F0	R2	0.445	0.465	0.997	0.962	0.955	73	69	4	0.568	-5.920	7.056
F0	R3	0.424	0.434	0.995	0.979	0.971	46	39	7	0.406	-6.755	7.567
F0	R4	0.512	0.531	0.993	0.955	0.949	83	72	11	0.505	-7.270	8.280
F0	R5	0.532	0.544	0.994	0.968	0.963	59	49	10	0.548	-10.472	11.567
F1	R1	0.423	0.437	0.996	0.973	0.965	56	51	5	0.643	-5.850	7.135
F1	R2	0.445	0.463	0.997	0.965	0.958	67	63	4	0.472	-5.447	6.392
F1	R3	0.424	0.432	0.996	0.983	0.976	38	32	6	0.330	-6.309	6.969
F1	R4	0.512	0.529	0.994	0.959	0.954	75	65	10	0.440	-7.039	7.919

filter	res	prev_ref	prev_pipe	sens	spec	kappa	flips	up	down	mean_diff	loa_lo	loa_hi
F1	R5	0.532	0.542	0.994	0.970	0.966	55	45	10	0.489	-10.309	11.288
F2	R1	0.423	0.435	0.996	0.976	0.968	51	45	6	0.554	-5.764	6.872
F2	R2	0.445	0.463	0.997	0.966	0.959	66	62	4	0.481	-5.389	6.352
F2	R3	0.424	0.430	0.995	0.986	0.979	34	27	7	0.361	-6.270	6.992
F2	R4	0.512	0.530	0.995	0.959	0.954	74	65	9	0.468	-7.003	7.939
F2	R5	0.532	0.543	0.995	0.970	0.966	55	46	9	0.522	-10.193	11.236
F3	R1	0.423	0.425	0.993	0.992	0.985	24	15	9	-0.119	-4.023	3.786
F3	R2	0.445	0.461	0.997	0.968	0.962	61	57	4	0.291	-3.688	4.270
F3	R3	0.424	0.429	0.996	0.987	0.981	30	24	6	0.184	-2.590	2.958
F3	R4	0.512	0.531	0.995	0.958	0.954	75	67	8	0.373	-5.929	6.674
F3	R5	0.532	0.543	0.995	0.970	0.967	54	45	9	0.413	-9.428	10.253
F4	R1	0.423	0.428	0.996	0.989	0.983	27	21	6	0.249	-4.599	5.098
F4	R2	0.445	0.464	0.997	0.965	0.958	68	64	4	0.478	-5.146	6.102
F4	R3	0.424	0.429	0.995	0.988	0.981	30	23	7	0.305	-5.850	6.460
F4	R4	0.512	0.533	0.995	0.953	0.949	82	74	8	0.471	-6.860	7.802
F4	R5	0.532	0.544	0.995	0.969	0.966	55	47	8	0.505	-9.917	10.927
F5	R1	0.423	0.428	0.996	0.989	0.983	27	21	6	0.248	-4.601	5.096
F5	R2	0.445	0.455	0.996	0.979	0.973	43	37	6	0.152	-5.131	5.436
F5	R3	0.424	0.428	0.995	0.988	0.982	29	22	7	0.290	-5.847	6.427
F5	R4	0.512	0.518	0.986	0.973	0.959	66	42	24	-0.051	-6.648	6.546
F5	R5	0.532	0.521	0.966	0.985	0.950	81	23	58	-0.322	-10.248	9.605
F6	R1	0.423	0.424	0.991	0.991	0.981	30	17	13	0.095	-4.794	4.984
F6	R2	0.445	0.361	0.806	0.996	0.818	287	7	280	-2.736	-12.533	7.062
F6	R3	0.424	0.362	0.846	0.995	0.857	222	10	212	-1.934	-11.830	7.961
F6	R4	0.512	0.203	0.395	0.999	0.388	1009	2	1007	-8.671	-32.380	15.038
F6	R5	0.532	0.110	0.207	0.999	0.195	1372	2	1370	-11.952	-47.877	23.973

flips: patients whose category differs (up: exposed by pipeline only). R4/R5 at phase offset 0. mean_diff and limits of agreement in minutes below 65 mmHg.

Table S3. POST HOC: probability that an episode is waveform-flagged, by elapsed time and nadir so far

still < 65 mmHg after (s)	nadir_so_far	episodes	p_flagged
10	<20	1406	0.747
10	20-39	2046	0.792
10	40-54	3301	0.478
10	55-64	33017	0.016
30	<20	384	0.594
30	20-39	704	0.574
30	40-54	2057	0.070
30	55-64	15900	0.009
60	<20	224	0.442
60	20-39	300	0.080
60	40-54	2018	0.017
60	55-64	9413	0.007
120	<20	159	0.289
120	20-39	248	0.024
120	40-54	1993	0.010
120	55-64	4911	0.006

Among episodes that had lasted at least the stated time; nadir so far = lowest monitor MAP up to that time.

Table S4. POST HOC: depth-weighted exposures and lower thresholds (1-min median resolution)

metric	thr	filter	res	or_ref	or_pipe	ror	lo	hi	p_perm
twa	55	F0	R3	3.893	1.186	0.305	0.061	1.531	0.105
twa	55	F1	R3	3.893	4.065	1.044	0.401	2.715	0.880
twa	55	F3	R3	3.893	4.767	1.224	0.546	2.744	0.470
twa	65	F0	R3	1.266	1.169	0.923	0.788	1.083	0.235
twa	65	F1	R3	1.266	1.273	1.005	0.953	1.060	0.780
twa	65	F3	R3	1.266	1.269	1.002	0.958	1.048	0.870
auc	55	F0	R3	3.990	2.413	0.605	0.197	1.853	0.335
auc	55	F1	R3	3.990	3.339	0.837	0.406	1.726	0.565
auc	55	F3	R3	3.990	3.812	0.955	0.653	1.398	0.730
auc	65	F0	R3	1.242	1.217	0.980	0.911	1.054	0.605
auc	65	F1	R3	1.242	1.242	1.000	0.968	1.033	0.995
auc	65	F3	R3	1.242	1.247	1.004	0.981	1.028	0.705

item	value
auc55_F0xR3_total_ratio	1.5926511038398004
auc55_F0xR3_excess_share	0.3721160914722307
auc55_F0xR3_top1pct_share_of_excess	0.5633518466768042
auc55_F1xR3_total_ratio	1.1474727971391918
auc55_F1xR3_excess_share	0.12851964552611783
auc55_F1xR3_top1pct_share_of_excess	0.5537390027342186
auc55_F3xR3_total_ratio	0.9909324504685909
auc55_F3xR3_excess_share	-0.009150522346019763
auc55_F3xR3_top1pct_share_of_excess	-2.0514721232838955
auc65_F0xR3_total_ratio	1.1250310833187709
auc65_F0xR3_excess_share	0.11113567009183167
auc65_F0xR3_top1pct_share_of_excess	0.5190926964251571
auc65_F1xR3_total_ratio	1.0485801543205124
auc65_F1xR3_excess_share	0.04632946191127629
auc65_F1xR3_top1pct_share_of_excess	0.5245433707005375
auc65_F3xR3_total_ratio	1.0101195139055774
auc65_F3xR3_excess_share	0.01001813524664108
auc65_F3xR3_top1pct_share_of_excess	0.7610090577627407
binary55_F0xR3	ROR 1.19 (0.66–2.15)
binary55_F1xR3	ROR 1.28 (0.81–2.02)
binary55_F3xR3	ROR 1.10 (0.84–1.45)
binary60_F0xR3	ROR 0.95 (0.79–1.14)
binary60_F1xR3	ROR 0.98 (0.85–1.15)
binary60_F3xR3	ROR 0.98 (0.84–1.13)

TWA per mmHg; AUC (area under the threshold) per 100 mmHg·min, capped at the reference 99th centile. excess_share: fraction of the pipeline total area in excess of the reference total; top1pct_share: share of that excess contributed by the 1% of patients with the largest excess. Intervals as in A15.

Table S5. Sensitivity analyses of the primary ROR (F1 × R3)

analysis	ror	lo	hi	or_pipe	or_ref	events
reference=unbridged	0.938	0.834	1.055			
reference=guard0	0.968	0.880	1.065			
reference=guard30	0.962	0.838	1.103			
reference=pp5	0.961	0.861	1.072			
reference=pp12	0.954	0.851	1.070			
reference=valid25	0.969	0.865	1.086			
reference=valid75	0.944	0.834	1.069			

analysis	ror	lo	hi	or_pipe	or_ref	events
reference=wavem.743		0.476	1.159			
S10 numerics-defined period	0.910	0.747	1.107			
S5 + blood loss	0.961			0.904	0.941	
S11 + number of postoperative creatinines	0.949			0.845	0.890	
S7 AKI stage >= 2 (reduced set)	0.968			2.143	2.215	24.000
S4 missing postoperative creatinine = no AKI (pre-specified)	0.963	0.865	1.071			91.000
S6 transplantation included (pre-specified)	0.958	0.866	1.060			122.000
without gynaecology (low creatinine testing)	0.959	0.857	1.073			89.000
disease risk score, 2 df	0.958	0.852	1.077			91.000
reference=concord	0.963	0.854	1.087			91.000
reference=valid25940s		0.850	1.060			91.000
reference=valid75952s		0.832	1.090			91.000

Rows from the frozen run first; S4, S6 (pre-specified, run after unblinding) and post hoc rows after. Rows without lo/hi are point estimates.

Table S6. Differential misclassification: AKI on spurious minutes per hour, adjusted for reference exposure

pipeline	model	term	nonzero	or_per_sd	lo	hi	or_per_log1p_u	lo_u	hi_u
F1xR3	b_g1_zero_c	high	19.000	1.022	0.829	1.259	1.892	0.004	966.903
F1xR3	b_g2_nonp	uls_noise	168.000	0.895	0.695	1.153	0.266	0.013	5.480
F1xR3	b_all	0	241.000	0.831	0.626	1.102	0.159	0.010	2.614
F1xR3	c_negative_c	control	241.000	0.832	0.628	1.103	0.161	0.010	2.641
F1xR3	c_negative_c	control	688.000	1.135	0.936	1.376	2.039	0.690	6.027
F1xR3	d_mechanism	0	241.000	0.837	0.632	1.107	0.170	0.011	2.740
F1xR3	d_mechanism	1	2.49e+03	0.978	0.772	1.239	0.940	0.492	1.797
F1xR3	d_mechanism	2	2.56e+03	1.188	0.938	1.504	1.197	0.936	1.530
F1xR3	d_mechanism	3	190.000	1.025	0.927	1.134	1.001	0.996	1.007
FoxR1	b_g1_zero_c	high	789.000	1.105	0.938	1.300	17.314	0.162	1.85e+03
FoxR1	b_g2_nonp	uls_noise	1.4e+03	0.878	0.695	1.110	0.288	0.031	2.710
FoxR1	b_all	0	1.73e+03	0.928	0.741	1.161	0.589	0.122	2.855
FoxR1	c_negative_c	control	1.73e+03	0.916	0.731	1.148	0.539	0.110	2.636
FoxR1	c_negative_c	control	1.94e+03	1.171	0.957	1.433	2.095	0.815	5.383
FoxR1	d_mechanism	0	1.73e+03	0.922	0.737	1.153	0.563	0.116	2.722
FoxR1	d_mechanism	1	2.49e+03	0.971	0.766	1.231	0.923	0.482	1.768
FoxR1	d_mechanism	2	2.56e+03	1.199	0.947	1.517	1.208	0.945	1.544
FoxR1	d_mechanism	3	190.000	1.026	0.928	1.135	1.001	0.996	1.007
FoxR1 (separate model)	spurious_hy	po		0.928	0.741	1.161	0.589	0.122	2.855
FoxR1 (separate model)	spurious_hy	per_negative_control		1.164	0.952	1.424	2.038	0.793	5.236

term 0: log(1 + spurious min/h) (or group-specific version); c_negative_control term 1: spurious hypertensive min/h; d_mechanism terms 1–3: blood gases/h, log blood loss, red-cell units. lo/hi per SD; lo_u/hi_u per unit of log(1 + min/h).

Table S7. Discordant patients, operating characteristics of the equivalence decision, and interval coverage

pipeline	up	aki_up	down	aki_down
F1xR3	28	0	5	0
FoxR1	55	4	1	0
F3xR2	50	3	4	0
F6xR3	8	0	178	3

F1 × R3: expected AKI among the 33 discordant patients 1.57; probability of none 0.19.

k	ror	lo	hi	n	events	verdict
0	0.956	0.856	1.068	2563	91	equivalent
1	1.013	0.901	1.138	2563	91	equivalent
2	1.070	0.900	1.272	2563	91	inconclusive
3	1.129	0.938	1.358	2563	91	inconclusive
4	1.195	0.957	1.492	2563	91	inconclusive
5	1.269	0.982	1.641	2563	91	inconclusive

ROR and A15 interval if k of the 28 upward-moved patients had developed AKI (outcomes swapped with k concordant cases; 400 bootstrap resamples, 200 permutations). Under the reference model, P(k = 0) = 0.26, P(k = 1) = 0.38, P(k ≥ 2) = 0.36.

scenario	gamma_up	median_ror	p_equivalent	coverage_of_1	sd_lror	n
null	0.000	1.004	0.820	0.970	0.062	200
upper_margin	1.912	1.239	0.050	0.510	0.098	200

Simulated outcomes from the fully adjusted reference model; differential error added as a log-odds shift γ in upward-moved patients. Each dataset analysed with the A15 rule (200 bootstrap resamples, 200 permutations). Median ROR by γ: up0.0: 0.99, up1.0: 1.08, up2.0: 1.27, up3.0: 1.52, up4.0: 1.89, down2.0: 0.95, down4.0: 0.86, down8.0: 0.78. Coverage of 1 under independent permutations (SD from the dry run, 100 calibration datasets): F1xR3 1.00, F0xR1 0.99, F3xR2 1.00, F6xR3 0.97.

Table S8. Injection–recovery sweep (post hoc)

kind	param	any_artefact	frac_artefact
flush	(120, 0.5)	0.000	0.000
flush	(120, 1)	0.000	0.000
flush	(120, 2)	1.000	0.100
flush	(120, 4)	1.000	0.200
flush	(150, 0.5)	0.000	0.000
flush	(150, 1)	0.000	0.000
flush	(150, 2)	1.000	0.100
flush	(150, 4)	1.000	0.200
flush	(200, 0.5)	0.000	0.000
flush	(200, 1)	0.000	0.000
flush	(200, 2)	1.000	0.100
flush	(200, 4)	1.000	0.200
flush	(250, 0.5)	1.000	0.100
flush	(250, 1)	1.000	0.100

kind	param	any_artefact	frac_artefact
flush	(250, 2)	1.000	0.100
flush	(250, 4)	1.000	0.200
flush	(300, 0.5)	1.000	0.100
flush	(300, 1)	1.000	0.100
flush	(300, 2)	1.000	0.100
flush	(300, 4)	1.000	0.200
zero_level	0	1.000	0.500
zero_level	5	1.000	0.500
zero_level	10	1.000	0.500
zero_level	15	1.000	0.500
zero_level	20	1.000	0.500
zero_level	25	1.000	0.500
damping_hz	1.5	0.118	0.079
damping_hz	2	0.008	0.001
damping_hz	3	0.000	0.000
damping_hz	5	0.000	0.000
damping_hz	10	0.000	0.000
partial_occlusion	0.5	0.000	0.000
partial_occlusion	0.3	0.025	0.010
partial_occlusion	0.15	0.613	0.520
drift_mmhg	10	0.008	0.001
drift_mmhg	20	0.008	0.001
drift_mmhg	40	0.008	0.006
ringing_mmhg	5	0.000	0.000
ringing_mmhg	10	0.000	0.000
ringing_mmhg	20	0.008	0.001
sampling_no_flush	mean	1.000	0.700
sampling_no_flush	20	1.000	0.700

any_artefact: share of injections with at least one window in the injected 20 s labelled artefact (zero/high/non-pulsatile/noise; guard excluded); 119 clean 2-min segments from 60 cases.

Table S9. Blinded AI rating of 152 snippets (secondary)

item	value
n_rated	152.000
n	131.000
agree	0.977
kappa	0.954

item	value
artefact_share_in_subset	0.435
alg_artefact_rated_artefact	1.000
alg_valid_rated_artefact	0.041
alg_valid_n	74.000
alg_valid_low_strata_n	36.000
guard_n	19.000
guard_rated_artefact	0.684
unsure	2.000

Table S10. In-hospital mortality (exploratory, reduced adjustment set)

n = 2962, deaths = 26; OR reference 1.50 (0.69–3.27); OR F1 × R3 1.47 (0.68–3.20); ROR 0.98.

Table S11. Odds ratios for AKI by adjustment level (AKI set, 1-min median)

series	exposure	adjustment	odds_ratio	lo	hi
reference	binary65	crude	1.655	1.090	2.513
reference	binary65	reduced_set	1.463	0.952	2.248
reference	binary65	full_set	0.940	0.596	1.483
reference	binary65	full_set_without_duration	1.097	0.704	1.709
reference	binary55	crude	3.813	1.865	7.794
reference	binary55	reduced_set	3.362	1.575	7.180
reference	binary55	full_set	1.954	0.894	4.270
reference	minutes65	crude	1.215	1.136	1.300
reference	minutes65	reduced_set	1.195	1.112	1.283
reference	minutes65	full_set	1.079	0.994	1.172
F1xR3	binary65	crude	1.593	1.049	2.419
F1xR3	binary65	reduced_set	1.414	0.921	2.170
F1xR3	binary65	full_set	0.899	0.571	1.417
F1xR3	binary55	crude	4.673	2.465	8.857
F1xR3	binary55	reduced_set	4.106	2.068	8.151
F1xR3	binary55	full_set	2.494	1.224	5.080
F1xR3	minutes65	crude	1.221	1.142	1.305
F1xR3	minutes65	reduced_set	1.200	1.119	1.288
F1xR3	minutes65	full_set	1.087	1.002	1.179

reduced_set: age and eGFR splines, ASA ≥ 3, emergency. Wald intervals from Firth fits. AKI-set inclusion by department: General surgery 85% of 1749, Gynecology 28% of 172, Thoracic surgery 96% of 996, Urology 97% of 59.

Supplementary figures

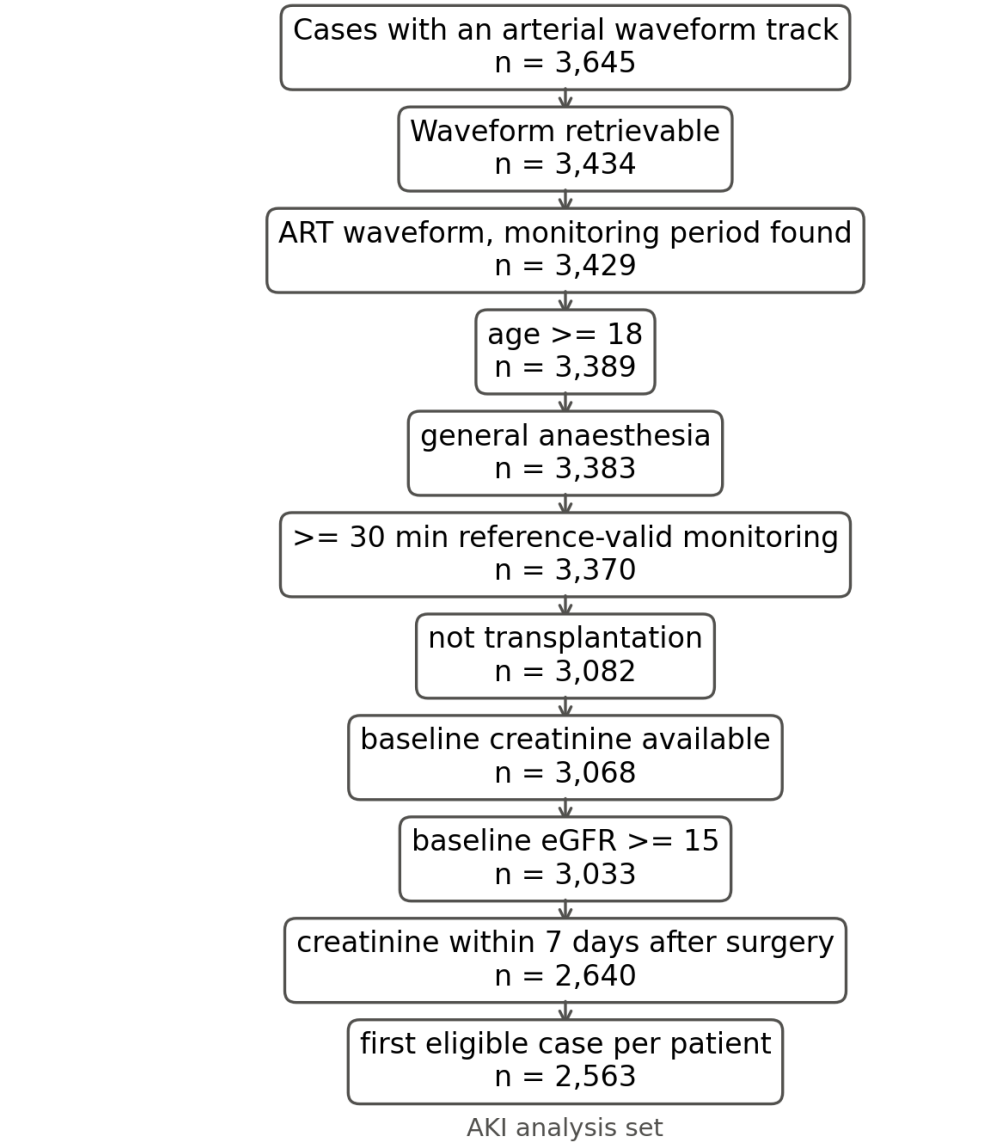


Figure S1. Flow of patients.

Supplementary Protocol

Analysis protocol (frozen before any outcome analysis)

Working title. How much of recorded intraoperative hypotension is artefact, and how much does it matter? A waveform-referenced multiverse analysis of arterial-line artefacts, data resolution and the hypotension–acute kidney injury association in VitalDB.

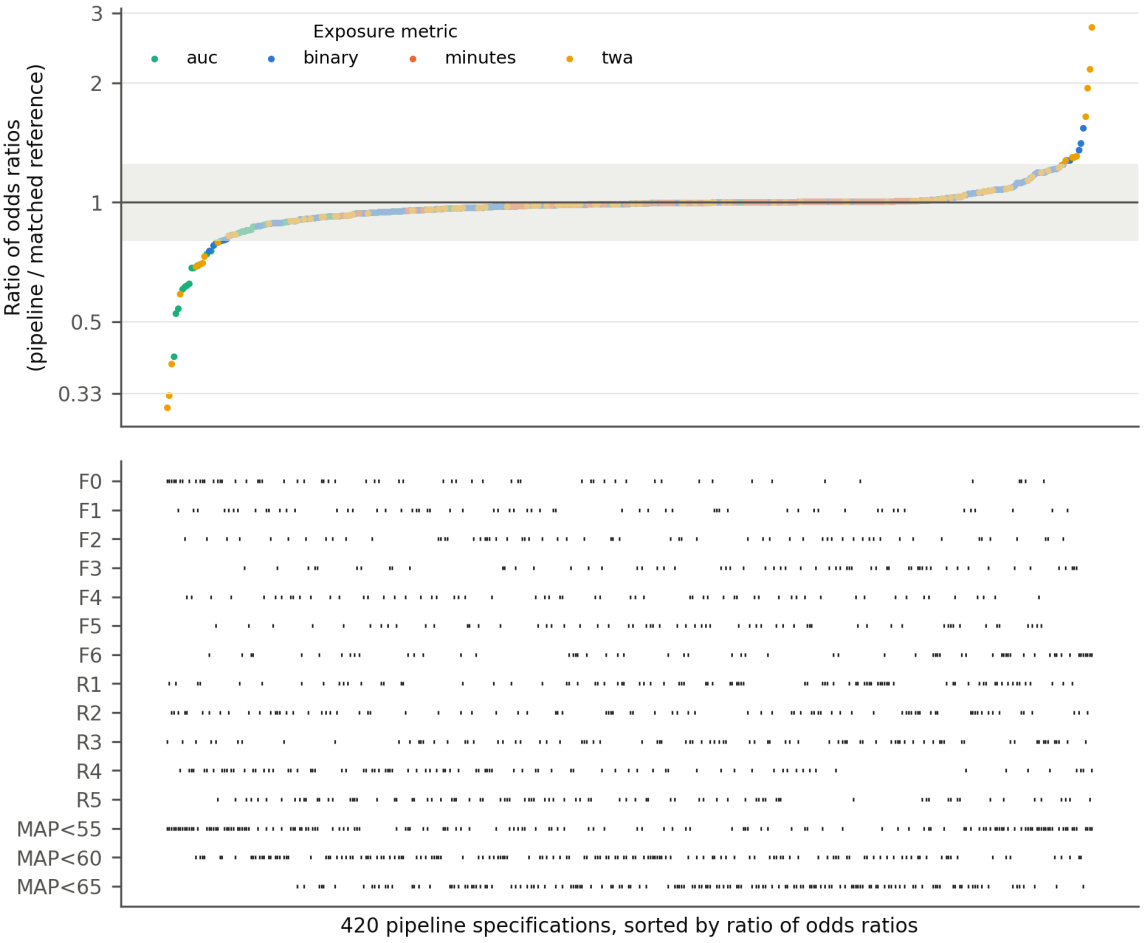


Figure S2. Specification curve of 420 pipeline specifications.

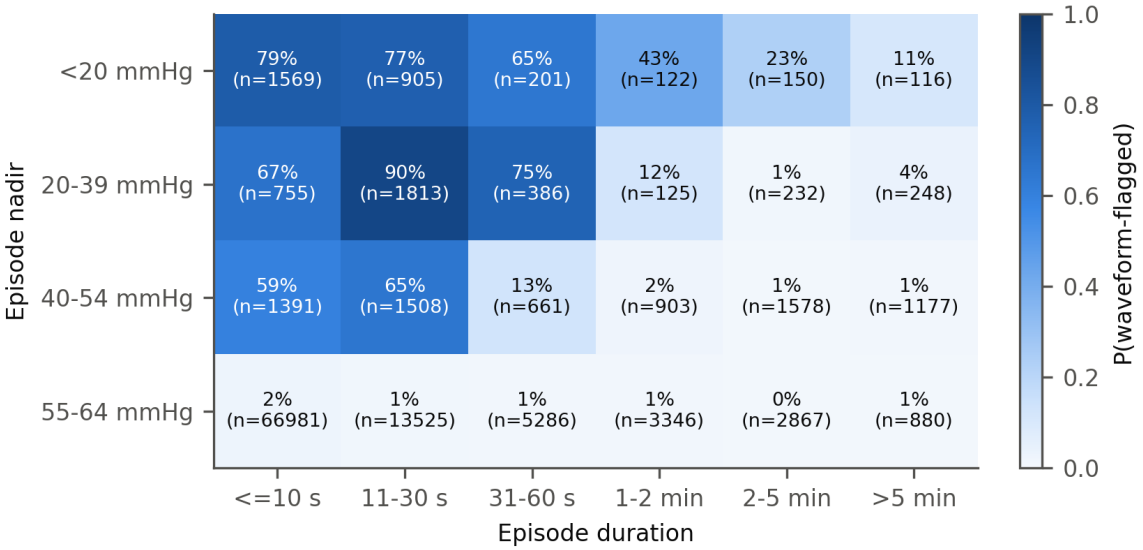


Figure S3. Probability that a recorded episode is waveform-flagged, by final duration and nadir.

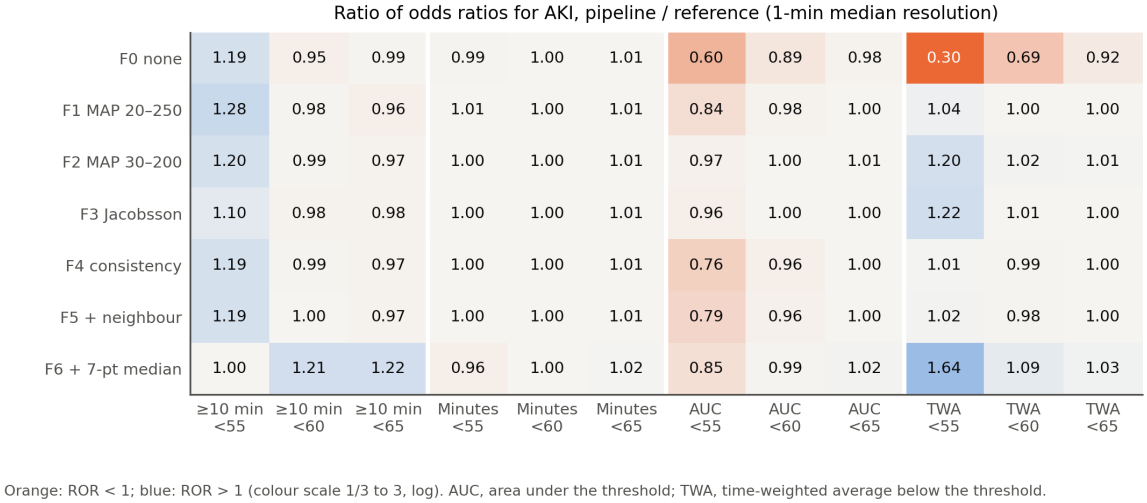


Figure S4. Ratio of odds ratios by filter, exposure metric and threshold (post hoc display).

Status at freezing. Window features extracted from the 500 Hz arterial waveform; artefact classifier inspected visually on 20 snippets and descriptive artefact burden computed on ≤ 600 cases. **No association between any exposure and any outcome has been computed.** The freeze is the git commit containing this file; any later change is recorded in the "Amendments" section with date and reason.

Reporting: STROBE and RECORD. Design: retrospective cohort, secondary analysis of a de-identified open dataset.

1. Objectives

#	Objective	Type
O1	Artefact burden of the intraoperative arterial line, by artefact type	descriptive
O2	Fraction of *recorded* hypotensive minutes that are artefact, for each processing pipeline	descriptive (primary measurement endpoint)
O3	Per-patient misclassification of hypotension exposure versus the reference	descriptive
O4	How the adjusted association between hypotension and AKI changes across pipelines (multiverse / specification curve)	**primary inferential**

#	Objective	Type
O5	Differential misclassification: is artefact-attributable "hypotension" associated with blood-gas sampling, severity and AKI?	secondary, hypothesis-driven
O6	Probability that a low monitor MAP episode is artefact, by duration and nadir; a simple bedside rule, derived and validated on separate patients	secondary, clinical

2. Data and cohort

VitalDB (Seoul National University Hospital, 2016–2017; Lee HC et al., Sci Data 2022). Open, de-identified; original collection approved by the SNUH IRB. No new ethics approval needed (to be confirmed against the VitalDB data-use terms and stated in the paper).

Inclusion: `SNUADC/ART` waveform present; age ≥ 18; general anaesthesia; ≥ 30 min of monitoring period (definition §3). Exclusions for the AKI analysis: missing baseline creatinine; baseline eGFR < 15 (proxy for end-stage renal disease); no creatinine within 7 days after the end of anaesthesia; transplantation procedures (`optype == 'Transplantation'`; included in sensitivity S6).

3. Reference series (waveform-informed)

- 2-s windows of the 500 Hz waveform are labelled `valid`, `missing`, `zero`, `high`, `nonpulsatile`, `noise` (rules in `src/artifact.py`, frozen at this commit), plus `guard`: windows within 10 s of any non-valid window.
- The monitor's 2-s ART_MBP numerics are aligned to the waveform by a per-case lag (1–4 samples, maximum correlation on valid windows).
- **Reference minute MAP** = median of the aligned monitor MAP values in `valid` windows of the minute, if ≥ 50% of the minute's windows are valid; otherwise missing.
- Rationale: the reference differs from the monitor stream only by removal of waveform-detected artefact, so calibration differences between the monitor and the ADC do not enter the comparison.
- **Monitoring period** = from the first to the last run of ≥ 60 s of valid windows within anaesthesia (anestart–aneend). All pipelines use the same period.

4. Pipelines (the multiverse)

Pipelines use **only the monitor numerics** (MAP, SBP, DBP), as a retrospective study would.

Filter (applied to 2-s numerics; F5 and F6 after sampling, see A8):

Code	Rule	Source
F0	none	—
F1	drop MAP < 20 or > 250	Lee 2026 (VitalDB AKI)
F2	drop MAP < 30 or > 200	Khan 2022
F3	drop SBP > 250, DBP < 15, MAP < 35 or > 150, SBP−DBP < 15	Jacobsson 2026
F4	drop MAP < 20 or > 200, SBP > 300 or < 20, SBP ≤ DBP, SBP−DBP < 5, DBP < 5	physiological consistency
F5	F4 + drop values differing from both neighbours by > 30 mmHg	neighbour comparison (Du 2019 style)
F6	F4 + 7-point running median on the minute grid	median filter (Pasma 2020)

Resolution: R1 2-s (all values; minutes = count × 2 s / 60); R2 1-min instantaneous (value at the minute mark); R3 1-min median; R4 5-min instantaneous, held; R5 15-min instantaneous, held.

Metric: cumulative minutes < T; area under T (mmHg·min); time-weighted average below T (AUC / monitored minutes); binary "≥ 10 cumulative minutes < T". **Threshold:** T in {55, 60, 65} mmHg.

7 × 5 × 4 × 3 = **420 pipeline specifications**, each compared with the reference computed with the same metric and threshold (12 reference specifications).

5. Outcomes and covariates

- **Primary:** AKI by KDIGO creatinine criteria within 7 days (≥ 0.3 mg/dL within 48 h, or ≥ 1.5× baseline within 7 days). Baseline = last creatinine within 30 days before induction, else `preop_cr`. Urine output not available postoperatively.
- **Secondary:** AKI stage ≥ 2; in-hospital death.
- **Adjustment set (fixed):** age, sex, BMI, ASA ≥ 3, emergency, baseline eGFR (CKD-EPI 2021), hypertension, diabetes, preoperative haemoglobin, preoperative albumin, department, surgical approach, log anaesthesia duration. Missing covariates: median (continuous) or mode (categorical) imputation with a missingness indicator if > 5% missing. Intraoperative vasopressors, fluids and blood loss are not adjusted for (potential mediators); sensitivity S5 adds blood loss.

6. Statistical analysis

- **O2.** Among pipeline-recorded minutes with MAP < 65: proportion where the reference minute is missing because of artefact, or ≥ 65. Reported per pipeline with patient-cluster bootstrap 95% CI (1,000 resamples).

- **O3.** For the binary exposure, sensitivity, specificity and Cohen's kappa against the reference. For minutes < 65, the median per-patient difference and Bland–Altman limits.
- **O4.**
  - For each of the 432 specifications, a logistic regression of AKI on the exposure plus the adjustment set.
  - Units:
    - binary: OR;
    - minutes: per 10 min, truncated at 120 min;
    - AUC: per 100 mmHg·min, truncated at the 99th percentile of the reference distribution;
    - TWA: per 1 mmHg.
  - **Primary estimand:** the ratio of odds ratios (ROR = OR_pipeline / OR_reference) for the binary exposure "≥ 10 min with MAP < 65", with patient-level paired bootstrap 95% CI (1,000 resamples, the same resample for both models).
  - Paired bootstrap for all 35 filter × resolution specifications of the primary metric. The other specifications are reported as point estimates in a specification curve.
  - Fragility summaries: range and IQR of ORs; proportion of specifications whose 95% CI excludes 1; the number of specifications that change the qualitative conclusion.
- **O5.**
  - Artefact-attributable minutes = minutes < 65 (F1, R3) – reference minutes < 65.
  - (a) Spearman correlation with the number of intraoperative arterial blood gases (VitalDB labs with a pH result during anaesthesia) and with ASA.
  - (b) Logistic regression of AKI on artefact-attributable minutes, adjusted for reference minutes and the adjustment set. Any association indicates differential misclassification.
- **O6.**
  - Episodes are contiguous runs of aligned 2-s monitor MAP < 65 within the monitoring period.
  - An episode is **artefactual** if ≥ 50% of its windows are non-valid (any label other than `valid`, including `guard`).
  - P(artefact) is tabulated by duration (≤ 10 s, 10–30 s, 30–60 s, 1–2 min, 2–5 min, > 5 min) × nadir (< 20, 20–39, 40–54, 55–64), with Wilson CIs.
  - A simple rule is derived on a 50% patient split (hash of caseid) and validated on the other 50%. The rule uses only features visible at the bedside: duration, nadir, and the maximum MAP in the preceding 30 s.
- **Mortality** (few events, about 40): Firth-penalised logistic regression adjusted for age, ASA ≥ 3, emergency and eGFR. Reported descriptively with the specification spread only.

7. Validation of the reference

- **V1 injection–recovery:**

  - Clean 2-min segments from 100 randomly selected cases.
  - Inject synthetic artefacts (Li, Mark & Clifford 2009 types): flush square wave to 300 mmHg, zeroing (0 mmHg), sampling (flat line at line pressure, then flush), overdamping (low-pass), high-frequency noise, and a 15 mmHg transducer offset.
  - Report the detection sensitivity per type. Specificity is measured on uninjected clean segments.
  - The transducer offset is expected to be undetectable by design (limitation).
- **V2 blinded adjudication:** 150 stratified random 40-s snippets, plotted without labels and adjudicated by an independent AI reviewer agent as valid or artefact. Report kappa. The adjudicator is not a human expert, and this is disclosed.
- **V3 physiological concordance:** |ART – NIBP| at NIBP measurement times, for valid versus non-valid windows; agreement of the waveform HR with the monitor ECG HR.

8. Sensitivity analyses

Code	Change
S1	guard 0 s and 30 s
S2	pulse-pressure threshold for non-pulsatile 5 and 12 mmHg
S3	reference minute validity threshold 25% and 75%
S4	missing postoperative creatinine treated as no AKI (cohort without the 7-day creatinine requirement)
S5	add intraoperative blood loss to the adjustment set
S6	include transplantation
S7	AKI stage ≥ 2
S8	reference built on the waveform mean instead of the monitor numerics

9. What this study cannot show

- It cannot detect artefacts that leave a normal-looking waveform, such as transducer height errors. The reference is an improved series, not ground truth.
- It cannot identify the clinical response to artefacts: VitalDB records vasopressor bolus totals only.
- The data come from a single centre in 2016–2017.
- The associations are observational.

Amendments (2026-10-26, before freezing, after an adversarial methods review; no outcome data examined)

Where an amendment conflicts with the text above, the amendment governs.

A1 (adapted) — validation without human adjudicators. Two blinded human anaesthesiologist adjudicators are not available for this version. Validation of the reference therefore rests on (i) physiological concordance with independent sensors (V3, now primary), (ii) injection–recovery (V1), and (iii) a blinded AI rater (V2, explicitly secondary and non-decisive). Throughout the paper, minutes are called **"waveform-flagged"** rather than "artefact" unless independent evidence supports the stronger term. A stratified 400-snippet adjudication set (strata A–D of the review: 150 monitor MAP < 65 & non-valid incl. 50 guard-only; 100 MAP < 65 & valid; 100 random valid; 50 random non-valid with MAP ≥ 65; at most two per patient, selected by hash of subjectid) is generated and released with blank forms, so that human adjudication can be added later.

A2 — freeze. The freeze is a git tag `protocol-freeze` on a commit whose code implements §4–§6 with these amendments, passes `tests/test_pipelines.py`, and has completed an end-to-end run with the AKI and death vectors replaced by a seeded permutation (outputs archived in `out/results_permuted/`). The SHA-256 of `src/*.py` is recorded in the tag message. An external registry (OSF) was not available; the local git history is the record. Descriptive artefact statistics before freezing used ≤ 600 cases.

A3 — primary estimand. ROR = OR_P / OR_ref for "≥ 10 cumulative minutes with MAP < 65", P = F1 × R3 (MAP 20–250 range filter, 1-min median), ref = bridged reference at R3. Paired bootstrap, 2,000 resamples of patients, percentile 95% CI. Decision rule with equivalence margin 0.80–1.25: CI entirely within -> "artefact handling does not materially change the association"; CI excludes 1 and point estimate outside the margin -> "material change"; otherwise "inconclusive". Under non-differential misclassification of plausible size the ROR is expected near 0.90–0.97, so this estimand detects differential bias large enough to matter. Key secondaries: (a) F0 × R1 vs ref@R1; (b) per-10-min log-OR for cumulative minutes < 65, F1 × R3 vs ref@R3, difference with bootstrap CI and attenuation factor β_P/β_ref.

A4 — gap bridging. Runs of non-valid windows ≤ 3 min bounded by ≥ 30 s of valid windows on both sides are linearly bridged between the median valid MAP of the 30 s before and after. The unbridged reference is a sensitivity analysis. Each pipeline-recorded minute < 65 is *spurious* (reference ≥ 65), *confirmed* (reference < 65, ≥ 50% observed), *bridged-low* (reference < 65, < 50% observed) or *indeterminate* (reference missing). Primary O2 quantity = spurious / all recorded minutes < 65, reported with indeterminate / all; indeterminate minutes are never counted as artefact.

A5 — missing waveform. `missing` is not an artefact and creates no guard. Minutes with > 50% `missing` windows are excluded from every pipeline and the reference (common support).

A6 — patients. Unit = patient (`subjectid`); first eligible case per patient (lowest caseid). Bootstrap and the O6 split are by patient.

A7 — O5. (a) Spearman of spurious minutes per monitored hour with arterial blood gases per hour

and ASA. (b) Firth logistic regression of AKI on log1p(spurious minutes/h), adjusted for reference minutes < 65 (linear + restricted cubic spline), log monitored hours and the §5 set; separately for minutes dominated by zero/high labels (g1) and non-pulsatile/noise labels (g2). (c) Negative-control exposure: spurious hypertensive minutes (pipeline MAP > 120 with reference ≤ 100), in the same model. (d) Mechanism model adding blood gases per hour, log1p(blood loss) and red-cell units.

A8 — resolution. The reference is sampled at every resolution and each pipeline is compared with ref@R; the resolution effect is OR(ref@R)/OR(ref@R3). R4 uses all 5 phase offsets and R5 offsets 0, 3, 6, 9, 12 min; log-ORs are averaged over offsets within each bootstrap resample. A charted value is the first non-missing 2-s value in its interval. F5 and F6 are applied after sampling on the pipeline's own grid (F6 = 7-point running median). S10: each pipeline uses a numerics-defined monitoring period (first to last 60-s run of monitor MAP 40–150 within anaesthesia), as a retrospective study would.

A9 — classifier calibration (descriptive data only). Noise = waveform minimum < −10 mmHg or median |second difference| ≥ 4 mmHg (the single-spike slope rule is dropped). Guard is asymmetric: 2 s before and 8 s after each artefact run (95th centile of monitor recovery time after zero/high/non-pulsatile runs = 8 s; 99th = 16 s). Sensitivity S1 uses 0/0 s and 6/30 s.

A10 — physiological checks (V3, now primary validation). (i) For monitor MAP < 65 episodes with an NIBP reading within ±3 min: proportion with NIBP MAP ≥ 70, by episode class; (ii) for non-pulsatile windows: proportion with concurrent ECG HR 40–150 and pleth HR 40–150; (iii) non-pulsatile fraction in patients with preoperative atrial fibrillation versus sinus rhythm. V1 specificity is reported as the flag rate outside the injected stretch.

A11 — O6. Probability that a monitor-numerics low-MAP episode is waveform-flagged, for retrospective cleaning and automated systems (not bedside use). Episode classes: *flagged* (≥ 50% artefact labels), *guard-dominated* (artefact + guard ≥ 50%, artefact < 50%), *physiological* (otherwise). Patient-cluster bootstrap CIs. Rule: depth-≤ 2 tree on duration, nadir and maximum MAP in the preceding 30 s, derived on a 50% patient-hash split with leaves called "flagged" only at ≥ 90% specificity; validated on the other half (sensitivity, specificity, PPV with cluster CIs).

A12 — models. Age and eGFR as restricted cubic splines (3 knots). Firth-penalised likelihood for all models; failures counted. Simultaneous 95% bands (bootstrap max-|z|) for the 35 F × R RORs. One analysis set for all specifications. S7 and mortality use a reduced set (age and eGFR splines, ASA ≥ 3, emergency). Mortality: one pre-specified specification (primary P vs ref) in the descriptive cohort, exploratory, no specification curve. S11 adds the number of postoperative creatinine measurements. Precision check before unblinding by simulation with assumed (not estimated) covariate effects and OR_ref = 1.5; if the expected 95% CI half-width of log ROR exceeds 0.22, the equivalence conclusion is declared unattainable in advance.

Cohorts. *Descriptive cohort* (O1–O3, O6, V3, mortality): adults, general anaesthesia, ≥ 30 min

reference-valid monitoring, first case per patient. *AKI analysis set* (O4, O5, S-analyses): additionally excludes transplantation, missing baseline creatinine, eGFR < 15, and no creatinine within 7 days.

A13 — classifier v2 (before freezing; no outcome data examined). The beat-lag autocorrelation pulsatility criterion was dropped: in descriptive data it labelled 12% of windows in patients with preoperative atrial fibrillation as non-pulsatile (1% in sinus rhythm), and those windows had pulse pressures of 56–83 mmHg; it also fired on bigeminy and slow heart rates. It was replaced by: pulse pressure < 8 mmHg, OR interdecile range < 3 mmHg, OR no systolic upstroke (rise > 8 mmHg within 100 ms) in the centred 6-s window. The noise rule became minimum < −10 mmHg OR median absolute second difference ≥ 4 mmHg (the earlier rule fired on single spikes). Guard is asymmetric: 2 s before and 8 s after an artefact run (95th centile of monitor recovery). All 3,645 waveform cases were re-extracted with these features; 211 had a waveform track listed but not retrievable and are excluded as "waveform not retrievable".

A14 — O5 power and Firth iterations (before freezing; only the permuted outcome examined). In the permuted dry run, spurious minutes at F1 × R3 were non-zero in 241 of 2,563 patients (zero/flush-dominated in 19), which cannot carry the differential-misclassification test. O5 is therefore run twice: at F1 × R3 as pre-specified, and at F0 × R1 (raw 2-s stream; non-zero in 1,728 patients), with odds ratios also reported per standard deviation of the exposure. The F0 × R1 version is the one interpreted. The Firth iteration limit was raised from 50 to 200: in the dry run 1.7% of bootstrap fits stopped at 50 iterations short of the 1e−7 tolerance, and refitting them changed the estimate only in the eighth decimal.

A15 — null calibration of the ROR interval (written before the calibration result was seen; real outcome never examined). In the dry run several RORs had percentile intervals excluding 1 although the permuted outcome makes the true ROR exactly 1, because the contrast rests on the few dozen patients whose exposure category differs. `src/null_calibration.py` re-permutes the outcome 100 times and, for F1 × R3, F0 × R1, F3 × R2 and F6 × R3, records how often the paired-bootstrap 95% interval (B = 200) covers 1, and the ratio of the mean bootstrap SE to the SD of log ROR across permutations. Decision rule, fixed now: - if coverage for F1 × R3 is ≥ 0.90, the percentile bootstrap interval stays primary; - otherwise every ROR interval (and simultaneous band) is computed as log ROR ± z × max(bootstrap SE, permutation-null SD of that cell), with the null SD estimated by 200 permutations per cell, and the percentile interval is reported as secondary. The equivalence verdict uses the primary interval.

A15 result (100 permutations, B = 200). Coverage of 1 by the percentile interval: F1 × R3 0.51, F0 × R1 0.79, F3 × R2 0.89, F6 × R3 0.91. The bootstrap distribution of log ROR is spiky — most resamples change few of the patients who drive the contrast — so its central 95% is narrow even where its SD is not (mean bootstrap SE / null SD 0.80 for F1 × R3). The pre-specified rule therefore applies: primary intervals are log ROR ± z × max(bootstrap SE, permutation-null SD), with 200 permutations per cell.

Under the same 100 null permutations this interval covered 1 in 0.99, 0.99, 1.00 and 0.97 respectively (conservative). The rule is applied to all ROR-type contrasts: the 35 F × R cells and their simultaneous bands, the resolution effect, the dose-slope differences and the reference-variant sensitivity analyses.

Notes after unblinding

Code hash. The `protocol-freeze` tag message records the SHA-256 printed by the final permuted dry run (12375fea…). Between that dry run and the freeze commit, only `src/figures.py` and `src/tables.py` (plotting and table formatting) were edited, so the hash of `src/*.py` at the tagged commit is 871bd97e…, which is the hash recorded by the real analysis (`out/results_real/summary.json`). `git diff protocol-freeze -- src/analysis.py src/models.py src/pipelines.py src/artifact.py` is empty for the real run.

Post hoc (not pre-specified). `src/posthoc_depth.py` adds A15-calibrated intervals for the depth-weighted RORs (AUC and TWA below 55 and 65 mmHg, F0/F1/F3 at R3), prompted by their point estimates in the specification curve. Reported as post hoc. The depth-metric heatmap (Figure 3) and forest plot (Figure 6) were added after unblinding.

Post hoc (not pre-specified). `src/posthoc_cuff_site.py` compares the peri-cuff artefact excess between radial and pedal/femoral catheters (an arm cuff cannot occlude the latter) and between right and left radial catheters, prompted by the C3 result.

Round-1 review changes (2026-10-26, after unblinding). Three AI peer reviewers (statistics, clinical anaesthesia, signal processing/reproducibility; `docs/reviews/`) led to: - *Cuff timing fixed.* `n_NIBP_MBP` had been shifted by the arterial-numerics lag (pipelines.py, figures.py). Cuff values are not delayed like the arterial numerics, so the shift was removed. This affects only C3, V3 and Figure 2 (outcome-free). The primary reference was re-checked after re-running `run_all.py`: OR_ref and OR_pipe for F1 × R3 are identical to the frozen run (0.9403 and 0.8992). - *Reference variants added (sensitivity only; primary unchanged).* `valid25_obs` and `valid75_obs` apply the minute-validity rule to observed valid windows only (the primary counts bridged values as valid by design of bridging, so the original valid25/valid75 barely tested the rule). `concord` also removes valid windows where monitor MAP and the waveform mean differ by > 10 mmHg. - *Pre-specified S4 and S6 had not been run.* They were omitted from `analysis.py` by error and are now run in `src/posthoc_stats.py` with the A15 interval. - *Post hoc analyses* in `src/posthoc_review.py`, `src/posthoc_models.py`, `src/posthoc_stats.py` and `src/validate_v1_sweep.py`: discordant-patient table, TOST operating characteristics by simulation, split-sample interval coverage, department restriction, disease-risk-score adjustment, crude-to-adjusted OR ladder, real-time episode table, pulse oximetry during deep episodes, cuff subtype/regularity/position, rule performance by nadir, injection sweep.

Round-2 review changes (2026-10-26). A coding error in the post hoc "without duration" model (`posthoc_models.py` fixed `ane_dur_h` instead of the `log_dur` column the model uses) was corrected;

the corrected estimate is 1.10 (0.70–1.71). The A11 rule is reported as the single split selected by the tree search. “Artefact” in the text denotes waveform-flagged values (A1). The fragility of the primary verdict to AKI among the discordant patients was quantified post hoc (`src/posthoc_k.py`).